LAB EXPERIMENT RECORD

22-10-2026---Set 27: Patient Health Risk Analysis

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Day	THURSDAY

Aim

To analyze patient health data in R and Tableau using a scatterplot matrix, Q-Q plot, ECDF, bar chart of averages and an interactive risk dashboard.

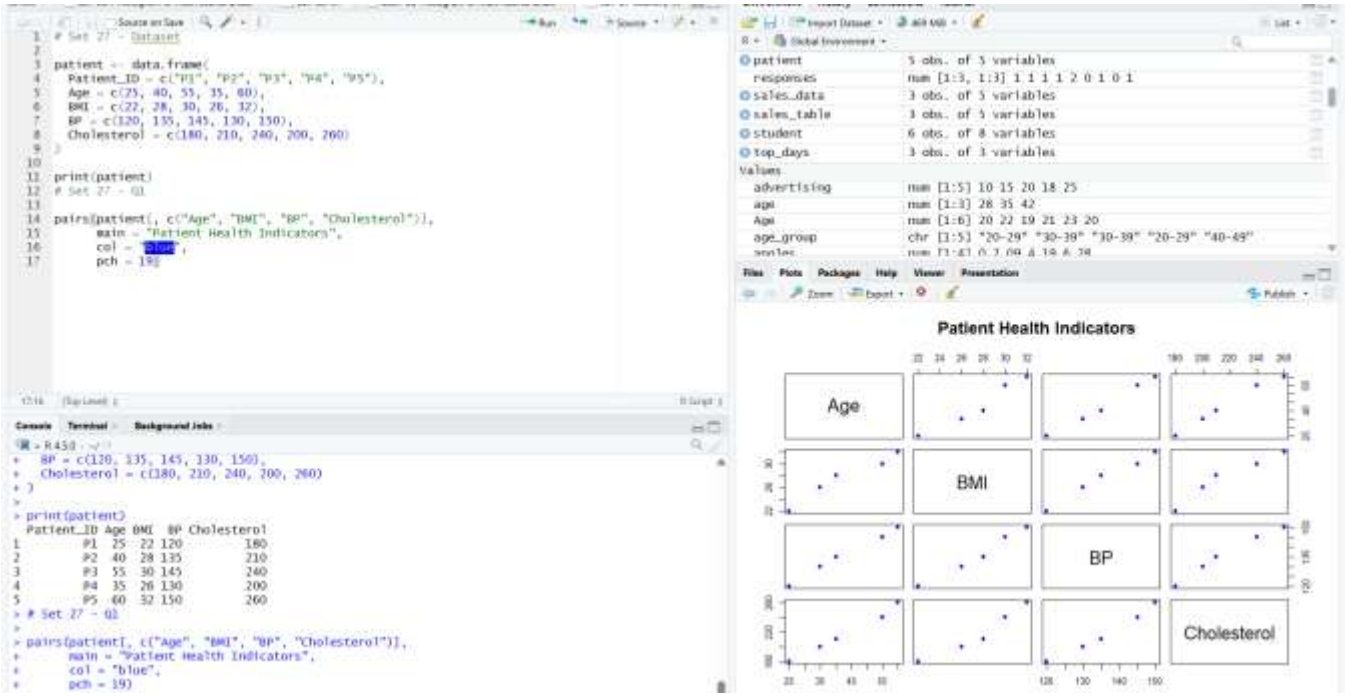
Dataset: Patient Health Risk

Patient ID	Age	BMI	BP	Cholesterol
P1	25	22	120	180
P2	40	28	135	210
P3	55	30	145	240
P4	35	26	130	200
P5	60	32	150	260

Question 1: Scatterplot Matrix of Health Indicators

Question: Using R, create a scatterplot matrix for Age, BMI, BP, and Cholesterol. Examine pairwise relationships to identify strong correlations, clusters, or potential risk patterns.

Aim: To examine all pairs of Age, BMI, BP and Cholesterol and to find strong relationships and risk patterns.



Result: All six pairs are strongly positive (correlations from 0.97 to 1.00), so older patients have higher BMI, BP and cholesterol, and P3 and P5 form a high-risk group while P1 sits alone at the low end.

Question 2: Q-Q Plot and ECDF of Cholesterol

Question: Construct Q-Q plot and ECDF for Cholesterol levels to assess distribution behaviour. Comment on normality assumption and skewness.

Aim: To check whether cholesterol levels follow a normal distribution using a Q-Q plot and an ECDF.

```
patient <- data.frame(
  Patient_ID = c("P1", "P2", "P3", "P4", "P5"),
  Age = c(25, 40, 55, 35, 60),
  BMI = c(22, 28, 30, 26, 32),
  BP = c(120, 135, 145, 130, 150),
  Cholesterol = c(180, 210, 240, 200, 260)
)

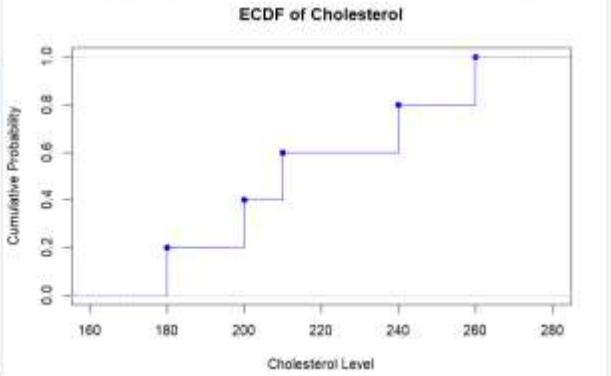
print(patient)
# Q-Q plot
qqnorm(patient$Cholesterol,
       main = "Q-Q Plot of Cholesterol",
       col = "blue",
       pch = 19)

qqline(patient$Cholesterol, col = "red")

# ECDF
plot(ecdf(patient$Cholesterol),
     main = "ECDF of Cholesterol",
     xlab = "Cholesterol Level",
     ylab = "Cumulative Probability",
     col = "blue",
     verticals = TRUE,
     do.points = TRUE)
```

```
5         P5  60  32 150         260
> # Q-Q plot
> qqnorm(patient$Cholesterol,
+        main = "Q-Q Plot of Cholesterol",
+        col = "blue",
+        pch = 19)
>
> qqline(patient$Cholesterol, col = "red")
>
> # ECDF
> plot(ecdf(patient$Cholesterol),
+      main = "ECDF of Cholesterol",
+      xlab = "Cholesterol Level",
+      ylab = "Cumulative Probability",
+      col = "blue",
+      verticals = TRUE,
+      do.points = TRUE)
>
```

```
patient          5 obs. of 5 variables
responses        num [1:3, 1:3] 1 1 1 1 2 0 1 0 1
sales_data       3 obs. of 5 variables
sales_table      3 obs. of 5 variables
student          6 obs. of 8 variables
top_days         3 obs. of 3 variables
Values
advertising      num [1:5] 10 15 20 18 25
age              num [1:3] 28 35 42
Age              num [1:6] 20 22 19 21 23 20
age_group        chr [1:5] "20-29" "30-39" "30-39" "20-29" "40-49"
```

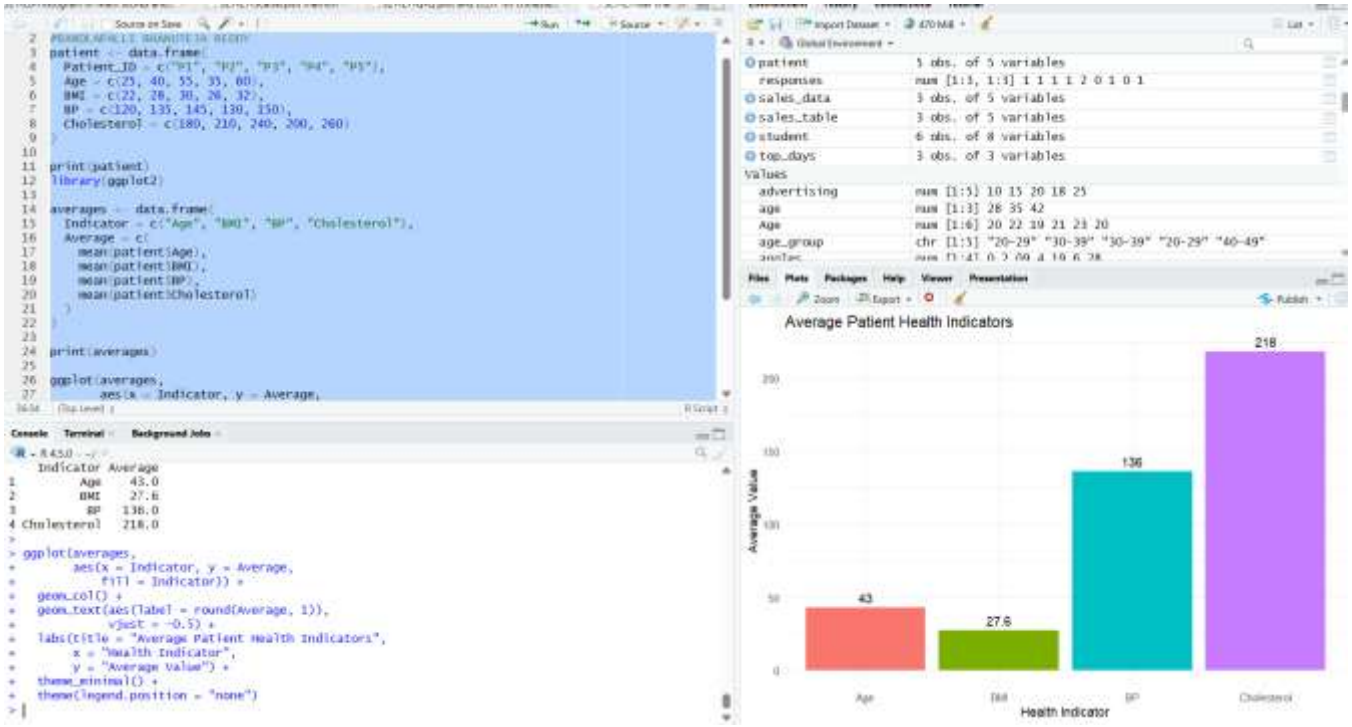


Result: The Q-Q points lie close to the straight line and the skewness is only about 0.2, so cholesterol looks roughly normal with a very slight right skew; with 5 patients this is only an indication.

Question 3: Bar Chart of Average Health Indicators

Question: Using R, create a bar chart showing the average values of each health indicator (Age, BMI, BP, Cholesterol). Include axis labels, a descriptive title, and distinct colors for each bar. How do these averages relate to general patient health risk?

Aim: To compare the average Age, BMI, BP and Cholesterol using a labelled bar chart with a different colour for each bar.



Result: The averages are Age 43, BMI 27.6, BP 136 and Cholesterol 218; BMI is in the overweight range, BP is above 130 and cholesterol is above 200, so the group is at raised health risk.

Question 4: Interactive Tableau Risk Dashboard

Question: Create an interactive Tableau dashboard with a heatmap highlighting high-risk patients and an Age filter for exploring risk patterns. Include hover tooltips showing all health indicators. Analyze which patients are highest risk, how age affects risk, and how this dashboard can aid healthcare professionals in prioritizing care.

Aim: To build an interactive Tableau heatmap dashboard with an Age filter and tooltips that highlights the patients at highest health risk.



Result: P5 (age 60, BMI 32, BP 150, cholesterol 260) and then P3 are the highest-risk patients, and risk rises steadily with age; the dashboard lets healthcare professionals spot the darkest cells quickly and prioritize those patients first.

Overall Result

Observation: The patients have an average age of 43, BMI 27.6, BP 136 and cholesterol 218, and every indicator rises with age, so P5 and P3 need care first.